

1.3.5 Satisfiability

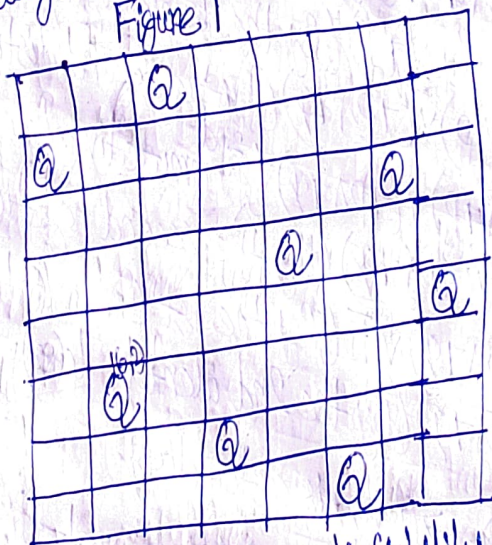
- A compound proposition is satisfiable if there is an assignment of truth values to its variables that makes it true (that is, when it is a tautology or a contingency).
- When no such assignment exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is unsatisfiable.

1.3.6 Applications of Satisfiability

Example 10: The n-Queens Problem

The n-queens problem asks for a placement of n queens on an $n \times n$ chessboard so that no queen can attack another queen. This means that no two queens can be placed in the same row, in the same column, or on the same diagonal. We display a solution to the eight-queens problem in Figure 1.

Figure 1



To model the n-queens problem as a satisfiability problem, we introduce n^2 variables, $p(i,j)$ for $i=1,2,\dots,n$ and $j=1,2,\dots,n$. For a given placement of queens on the chessboard, $p(i,j)$ is true when there is a queen on the square in the i th row and j th column, and is false otherwise. Note that squares (i,j) and (i',j') are on the same diagonal if either $i-i'=j-j'$ or $i-i'=j'-j$. In the chessboard in Figure 1, $p(6,2)$ and $p(2,1)$ are true, while $p(3,4)$ and $p(5,4)$ are false.

For no two of the n queens to be in the same row, there must be one queen in each row. We can show that there is one queen in each row by verifying that each every row contains at least one queen and that every row contains at most one queen.